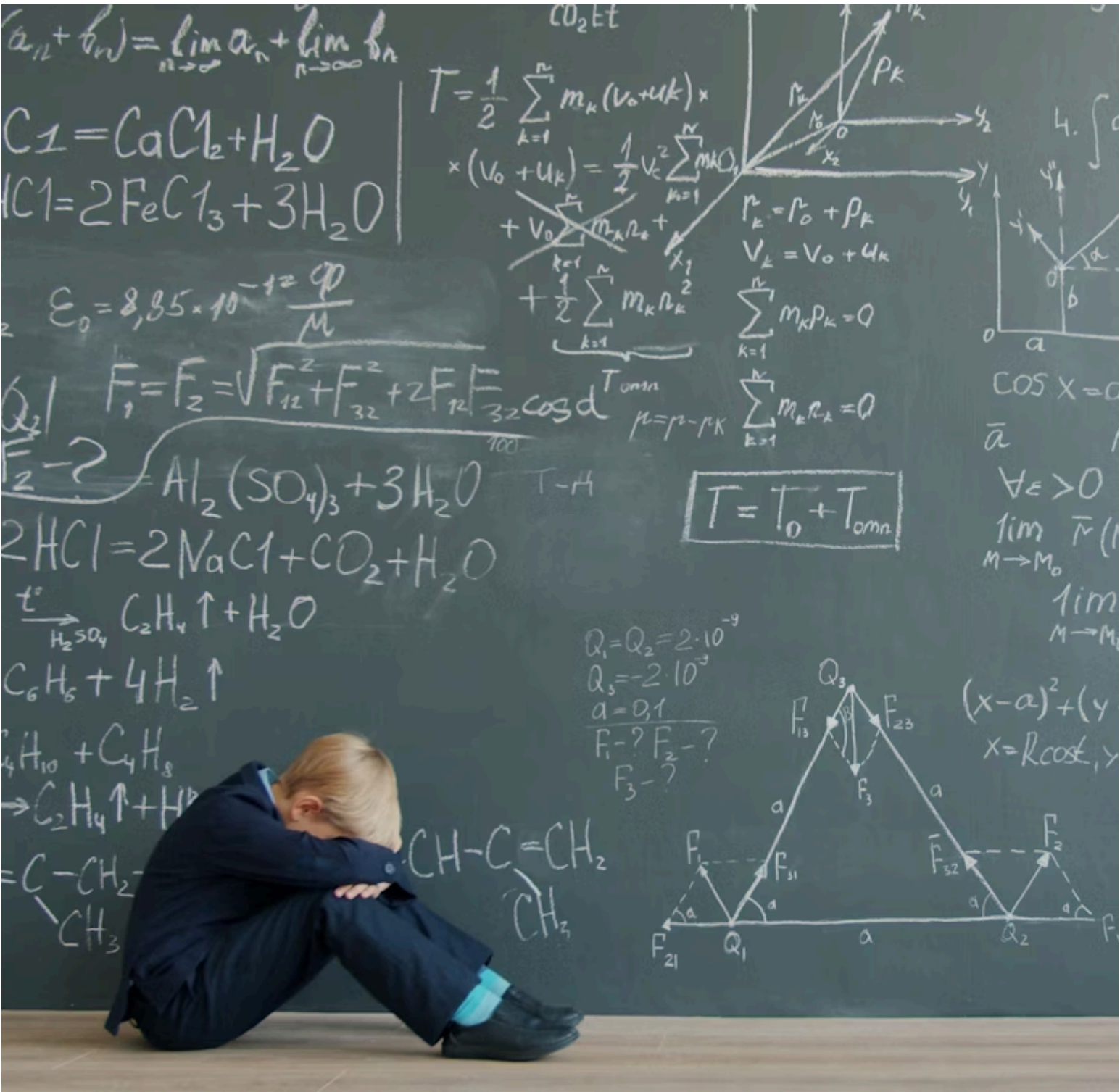


PROJECT 4

STUDENT DEPRESSION

A Statistical Analysis Using Two-Sample Z-Test for Proportions



BACKGROUND

Student mental health has become a growing concern worldwide. Family environment is believed to be one of the key factors influencing students' emotional well-being.

In this project, we analyze the Student Depression Dataset to investigate whether there is a significant difference in depression rates between students with and without a family history of mental illness. The goal is to generate insights that can help educational institutions provide better mental health support for students.

EXPLORATORY DATA ANALYSIS

5000

Data Sample

2583

No Family History

2583

With Family History

0

Missing/Duplicate Value

DEPRESSION	0	1
FAMILY HISTORY OF MENTAL ILLNESS		
NO	1116 (56.79%)	1467 (60.94%)
YES	944 (43.21%)	1473 (39.06%)

HYPOTHESIS TESTING

H_0 (Null Hypothesis)

$$H_0 : P_1 = P_2$$

There is NO significant difference in the proportion of depression between students with and without a family history of mental illness.

H_1 (Alternative Hypothesis)

$$H_1 : P_1 \neq P_2$$

There IS a significant difference in the proportion of depression between students with and without a family history of mental illness.

TEST PARAMETERS

Test Type	: Two-Sample Z-Test for Proportions
Significance Level	: $\alpha = 0.05$
Direction	: Two-Tailed Test
Critical Value	: $z_{\text{crit}} = \pm 1.96$

STATISTICAL TEST RESULT

Z-Statistic

2.9786

Computed test stat

Critical Value (z_crit)

±1.9600

Two-tailed, $\alpha=0.05$

P-Value

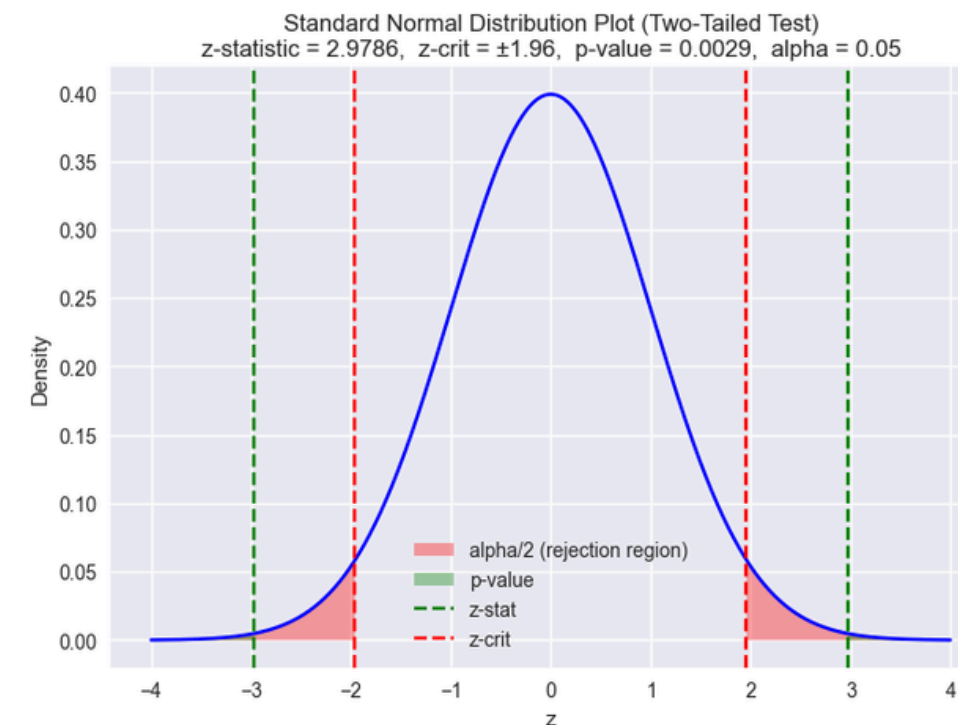
0.0029

$< \alpha = 0.05$

RESULT

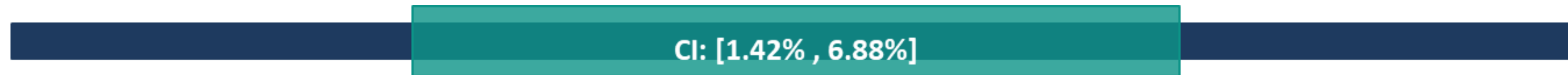
REJECT H_0

- Z-Statistic (2.9786) > Z-Critical (± 1.96)
- P-Value (0.0029) < α (0.05)



CONFIDENCE INTERVAL

Difference in Depression Proportion ($P_1 - P_2$)



- The difference in depression proportion ($P_1 - P_2$) falls within 1.42% to 6.88% at 95% confidence level.
- The entire interval is positive and does not include 0, confirming the difference is statistically significant.
- Students with family history: 60.94% depression rate vs. 56.79% for those without — a meaningful gap.

CONCLUSION

There is a statistically significant difference in depression rates between students with a family history of mental illness (60.94%) and those without (56.79%). The Two-Sample Z-Test ($z = 2.9786$, $p = 0.0029$) confirms we reject H_0 .

RECOMMENDATION

1

Priority Screening

Educational institutions should implement psychological screening programs targeting students with a known family history of mental illness.

2

Early Intervention

Provide proactive mental health support and counseling for high-risk students before depression symptoms worsen.

3

Awareness Programs

Develop campus-wide programs to reduce stigma around mental health and encourage students to seek help early.

**THANK YOU FOR
YOUR TIME**



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DATA ANALYST PRESENTATION
